

Lecture #5, Sep 23, 2025

One step time integration and amplification factor

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \quad x \in \mathbb{R}, \quad t > 0$$

$$u(x, 0) = f(x)$$

Re write. $\frac{\partial u}{\partial t} = L(u) \quad \left| \quad \frac{\partial u}{\partial t} = f(u) \right.$

$L(\cdot)$ is a function or a differential operator.

e.g. $L(u) = \frac{\partial^2 u}{\partial x^2} \quad : \text{diff. operator}$

$L(u) = \lambda u \quad : \text{function, } \lambda \text{ is scalar.}$

$$\text{let } u(x, t_n + \Delta t) = u(x, t_n) + \Delta t \left. \frac{\partial u}{\partial t} \right|_n + \frac{\Delta t^2}{2!} \left. \frac{\partial^2 u}{\partial t^2} \right|_n + \dots$$

denote $u(x, t_n) = u^n$

$$\frac{u^{n+1} - u^n}{\Delta t} = \left. \frac{\partial u}{\partial t} \right|_n + \text{error} = L(u^n)$$

$$\frac{u^{n+1} - u^n}{\Delta t} = L(u^n) \quad \textcircled{*}$$

$t_n \leftrightarrow n$: present time level or n -th time level

$t_n + \Delta t \leftrightarrow n+1$: future time level

$\textcircled{*}$ is called Euler explicit (Euler forward) method.

$$\textcircled{*} \Rightarrow u^{n+1} = u^n + \Delta t L(u^n)$$

$$u^{n+1}(x) = u^n(x) + \Delta t L(u^n(x))$$

Implementation (time marching method)

At $n=0$, $u^0 = u(x, 0) = \text{initial condition}$

At $n=1$, $u^1 = u^0 + \Delta t L(u^0)$, L is given

At $n=2$, $u^2 = u^1 + \Delta t L(u^1)$ u^1 is given.

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Euler Implicit

$$\frac{u^{n+1} - u^n}{\Delta t} = L(u^{n+1}), \quad u \text{ is the desired sol}^n$$

$$u^{n+1} = u^n + \Delta t L(u^{n+1})$$

$$u^{n+1} - \Delta t L(u^{n+1}) = u^n$$

$$\text{let } g(u^{n+1}) = u^n$$

$\Rightarrow$ linear eqⁿ if $g(\cdot)$ is linear

$\Rightarrow$ non-linear eqⁿ if $g(\cdot)$ is non-linear.

Remarks

$Ax = b$ is linear, $x = A^{-1}b$

$F(x) = b$ is non-linear. use Newton's method.

At $n=0$, $u^0 = u(x, 0)$ given

At $n=1$, solve $g(u^1) = u^0$ to get u^1
and so on.

Trapezoidal

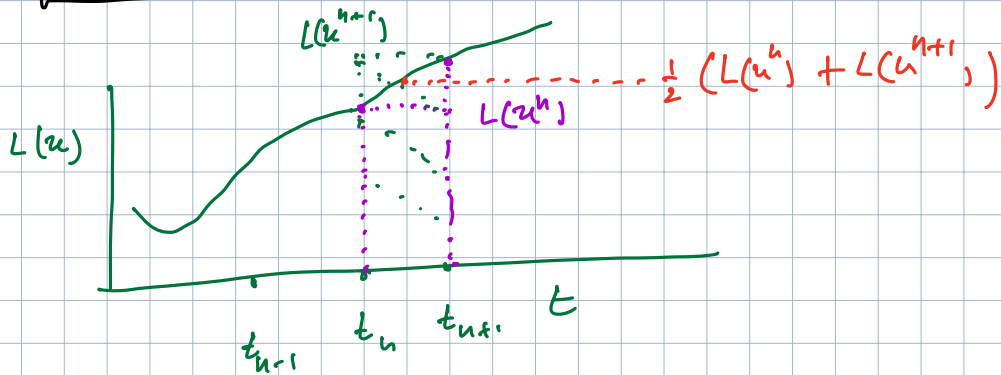
$$\frac{u^{n+1} - u^n}{\Delta t} = \frac{1}{2} [L(u^n) + L(u^{n+1})]$$

$$u^{n+1} = u^n + \frac{\Delta t}{2} L(u^n) + \frac{\Delta t}{2} L(u^{n+1})$$

$$u^{n+1} - \frac{\Delta t}{2} L(u^{n+1}) = u^n + \frac{\Delta t}{2} L(u^n)$$

$$\boxed{g(u^{n+1}) = u^n + \frac{\Delta t}{2} L(u^n)}$$

Graphical:



Errors:

Euler explicit $\Rightarrow O(\Delta t) \Rightarrow E(\Delta t) = C \Delta t$

Euler implicit $\Rightarrow O(\Delta t) \Rightarrow E(\Delta t) = C \Delta t$

Trapezoidal $\Rightarrow O(\Delta t^2) \Rightarrow E(\Delta t) = C \Delta t^2$

Convergence

let $L(\cdot)$ denote a function and satisfies:

there exist λ such that

$$\|L(u) - L(v)\| \leq \lambda \|u - v\|$$

for every u and v

Then Euler method converges
specifically

$$|u^n - u(x, t_n)| \leq \frac{C}{\lambda} \Delta t [(1 + \lambda \Delta t)^n - 1]$$

$\nwarrow$ numerical solⁿ $\swarrow$ Exact for some C , and $u \geq 0$

Ex: Heat eqⁿ with Euler Explicit method

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

$$\text{let } u_j^n = u(x_j, t_n), \quad j = 1, 2, \dots, N$$

$$n = 0, 1, \dots$$

$$\frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} = \frac{u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n))}{\Delta x^2}$$

$$\boxed{\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}}$$

Euler Explicit scheme

$$u_j^{n+1} = u_j^n + \frac{\Delta t}{\Delta x^2} (u_{j+1}^n - 2u_j^n + u_{j-1}^n) \quad \leftarrow \text{marching form}$$

H.W. Write Euler explicit scheme for

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0$$

using central scheme in space.

E.E. scheme for Heat

$$\frac{\partial u}{\partial t} = L(u) \equiv \frac{\partial^2 u}{\partial x^2}$$

$$\frac{u^{n+1} - u^n}{\Delta t} = L(u^n)$$

$$L(u) = \frac{\partial^2 u}{\partial x^2}$$

$$L(u^n) = L(u(x, t_n)) = \frac{u_{j+1}(t_n) - 2u_j(t_n) + u_{j-1}(t_n)}{\Delta x^2}$$

H.W. Write trapezoidal scheme for Heat eqⁿ using second order in space.

Stability:

Observe $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$

$\Rightarrow \frac{\partial \hat{u}}{\partial t} = -k^2 \hat{u}$, $\hat{u}$ is Fourier mode.

semi discrete Heat eqⁿ.

Let $u_j(t) = u(x_j, t)$, $x_j = j\Delta x$,

Then $\frac{\partial u_j}{\partial t} = \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}$

Recall dispersion relation (Week 2)

let $u(x_j, t) = e^{i(kx_j + \omega t)}$

Substitute on the right hand side

$$\frac{\partial u_j}{\partial t} = \left(-\frac{4}{\Delta x^2} \sin^2 \left(\frac{k\Delta x}{2} \right) \right) u_j$$

Thus we can write $L(u) = \lambda u$

$$\boxed{\frac{\partial u}{\partial t} = \lambda u}$$

Observe Euler explicit :

$$\frac{u^{n+1} - u^n}{\Delta t} = \lambda u^n \equiv L(u^n)$$

$$u^{n+1} = u^n + \lambda \Delta t u^n = (1 + \lambda \Delta t) u^n$$

$$\frac{u^{n+1}}{u^n} = 1 + \lambda \Delta t$$

$$\text{If } \left| \frac{u^{n+1}}{u^n} \right| = |1 + \lambda \Delta t| < 1$$

the solution is stable.

It is called A-stability.

Defin $G = \frac{u^{n+1}}{u^n}$ to be amplification factor.

Stability region is define by G on the
Complex plane $\text{Re } \lambda \Delta t$ vs $\text{Im } \lambda \Delta t$

Lecture #6, Sep 25, 2025

Ex: Derive stability region for Euler explicit method.

Consider $\frac{\partial u}{\partial t} = \lambda u$

$$\frac{u^{n+1} - u^n}{\Delta t} = \lambda u$$

For E.E., $\frac{u^{n+1}}{u^n} = (1 + \lambda \Delta t)$

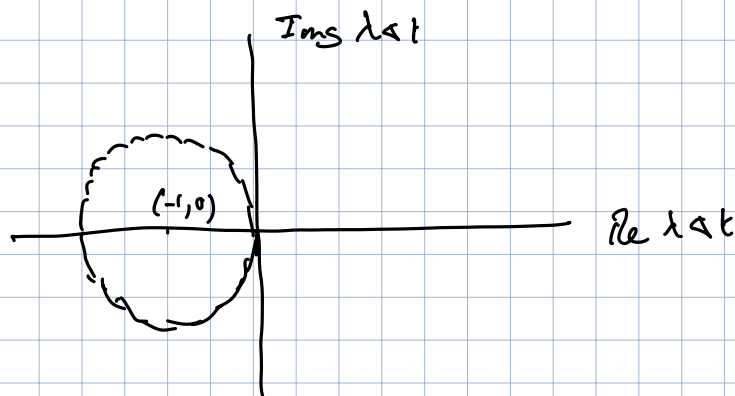
Stability region is the part of the complex plane $\text{Re } \lambda \Delta t$ vs $\text{Im } \lambda \Delta t$ which contains $|1 + \lambda \Delta t|$ i.e. the amplification factor.

let $z = \lambda \Delta t = x + iy$

$$1 + \lambda \Delta t = 1 + x + iy$$

$$|1 + \lambda \Delta t| = \sqrt{(1+x)^2 + y^2} = \sqrt{(1 + [\text{Re } \lambda \Delta t])^2 + [\text{Im } \lambda \Delta t]^2}$$

Region is given by $\sqrt{(1+x)^2 + y^2} = 1$



$$-1 < 1 + \lambda \Delta t < 1$$

$$-2 < \lambda \Delta t < 0 \Rightarrow \Delta t \leq \frac{2}{|\lambda|}$$

For a system: $\Delta t \leq \frac{2}{|\lambda_{\max}|}$

H.W Derive and sketch stability region for Euler Implicit and Trapezoidal methods.

Euler Implicit

$$G(z) \equiv G(\lambda \Delta t) = \frac{u^{n+1}}{u^n} = \frac{1}{(1 - \lambda \Delta t)}$$

Remark

Conditional stability means stability region is restricted.

Unconditional stability $\Rightarrow$ region is the entire left of complex plane

Heat eqⁿ (stability region)

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}, \quad \nu > 0$$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \nu \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}$$

Where $u_j^n = u(x_j, t_n)$

$$u_j^{n+1} = u_j^n + \frac{\nu \Delta t}{\Delta x^2} (u_{j+1}^n - 2u_j^n + u_{j-1}^n)$$

use $r = \frac{\nu \Delta t}{\Delta x^2}$

Plug $u_j^n = \hat{u}^n e^{ikx_j} \Leftarrow e^{i(kx + \omega t)} = e^{i\omega t_n} e^{ikx_j}$
 $= \hat{u}^n e^{ikx_j}$

into heat eqⁿ:

$$\hat{u}^{n+1} e^{ikx_j} = \hat{u}^n e^{ikx_j} + r \left(\hat{u}^n e^{ik(x_j + \Delta x)} - 2\hat{u}^n e^{ikx_j} + \hat{u}^n e^{ik(x_j - \Delta x)} \right)$$

$$\hat{u}^{n+1} = \hat{u}^n + r \hat{u}^n (e^{ik\Delta x} - 2 + e^{-ik\Delta x})$$

$$\frac{\hat{u}^{n+1}}{\hat{u}^n} = 1 + r(2\cos k\Delta x - 2)$$

$$G = 1 + 2r \left(-2 \sin^2 \frac{k\Delta x}{2} \right)$$

$$G(k) = 1 - 4r \sin^2 \frac{k\Delta x}{2}$$

Require $|G| \leq 1 \Rightarrow |1 - 4r \sin^2 \frac{k\Delta x}{2}| \leq 1$

$$-1 \leq 1 - 4r \leq 1$$

$$-2 \leq -4r$$

$$\Rightarrow r \leq \frac{1}{2}$$

$$\frac{2\Delta t}{\Delta x^2} \leq \frac{1}{2}$$

$$\Delta t \leq \frac{1}{2} \frac{\Delta x^2}{2}$$

Remark

$$\lambda = -\frac{4\nu}{\Delta x^2} \sin^2 \frac{k\Delta x}{2}$$

for heat eqⁿ (EE)

ODE method $\Rightarrow \Delta t \leq \frac{2}{|\lambda|}$

$$\lambda \in \left(-\frac{4\nu}{\Delta x^2}, 0 \right)$$

$$\Delta t \leq \frac{2}{4\nu/\Delta x^2} = \frac{\Delta x^2}{2\nu}$$

Numerical ex:

$$\frac{\partial u}{\partial t} = A u \quad \text{or} \quad \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -0.5 & 20 \\ 0 & -20 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

H.W. Write a python function that solve
a system of ODE using Euler method.

3. [Week 3]

(a) Consider the Initial Value Problem

$$\begin{aligned}u' &= 2u, \\v' &= 3u - v,\end{aligned}$$

with initial conditions specified at time $t = 0$. Solve this problem in two different ways:

- i. Solve the first equation, which only involves $u(t)$, and then insert this function into the second equation to obtain a nonhomogeneous linear equation for u_2 . Solve this to obtain $v(t)$. Plot the solutions $u(t)$ and $v(t)$. Choose any initial condition and numerical method. Compare the numerical solution with analytical solution.
 - ii. Write the given system as $\mathbf{u}' = A\mathbf{u}$. Draw the stability region for Euler explicit method overlaying the maximum acceptable time step. Compare the numerical solutions obtained by Euler Implicit and Trapezoidal method. Plot error $e(t)$ to observe the effect of reducing Δt .
- (b) Apply the trapezoidal method to the equation $u' = \lambda u$ and show that

$$u^{n+1} = G(z)u^n,$$

where $u^n = u(t_n)$, $z = \lambda\Delta t$, and

$$G(z) = \left(\frac{1 + z/2}{1 - z/2} \right).$$

- (c) Show that $G(z) = e^z + \mathcal{O}(z^3)$ and interpret why the local one-step error of the trapezoidal method on this problem is $\mathcal{O}(\Delta t^3)$ even though the method is second order accurate.

Hint: One way to obtain this result is to consider the expansion

$$\frac{1}{1 - z/2} = 1 + \frac{z}{2} + \left(\frac{z}{2}\right)^2 + \left(\frac{z}{2}\right)^3 + \dots$$

- (d) Solve the heat equation

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}, \quad x \in [a, b], \quad t > 0$$

subject to the boundary condition $u(a, t) = 0$ and $u(b, t) = 0$.

Write a code applying the trapezoidal method for solving the heat equation. Consider an initial condition with a narrow Gaussian profile. Compare the solution by plotting $u(x, T)$ and error $e(x, T)$ for a few values of Δt , which keeping Δx fixed. Discuss the behaviour of the solution if you increase or decrease Δt .

Hint: Modify the posted code to implement Trapezoidal method for heat equation.